PGCP-AC AUGUST 26
Database Assignment 1

Schema Design & Table Creation (DDL & DML Commands)

1. Define a schema for a Library Management System with the following entities:
   - Books
   - Authors
   - Members
   - Borrow_Records

```
mysql> create database lms
    -> ;
Query OK, 1 row affected (0.02 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| cdac               |
| information_schema |
| lms                |
| mysql              |
| performance_schema |
| sys                |
| temp               |
+--------------------+
7 rows in set (0.00 sec)
```

2. Write the SQL command to create a table Authors with the following fields:
   - author_id (Primary Key, INT)
   - name (VARCHAR(100))
   - country (VARCHAR(50))

```
mysql> create table Authors(author_id int primary key,name varchar(100),country varchar(50));
Query OK, 0 rows affected (0.04 sec)

mysql> desc Authors;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| author_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(100) | YES  |     | NULL    |       |
| country   | varchar(50)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
3 rows in set (0.02 sec)
```

3. Write the SQL command to create a table Books with the following fields:
   - book_id (Primary Key, INT)
   - title (VARCHAR(150))
   - author_id (Foreign Key referencing Authors)
   - published_year (YEAR)
   - available_copies (INT)

```
mysql> create table Books(book_id int primary key, title varchar(150),author_id int, published_year year,available_copies int, constr
aint fk_author_id foreign key (author_id) references Authors(author_id) on delete cascade);
Query OK, 0 rows affected (0.06 sec)

mysql> desc Books
    -> ;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| author_id        | int          | YES  | MUL | NULL    |       |
| published_year   | year         | YES  |     | NULL    |       |
| available_copies | int          | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

4. Write the SQL command to create a table Members with:
   - member_id (Primary Key, INT)
   - name (VARCHAR(100))
   - email (VARCHAR(100), unique)
   - phone (VARCHAR(15))

```
      create table Members(
    -> member_id int primary key,
    -> name varchar(100),
    -> email varchar(100) unique,
    -> phone varchar(15));
Query OK, 0 rows affected (0.04 sec)

mysql> desc Members;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| member_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(100) | YES  |     | NULL    |       |
| email     | varchar(100) | YES  | UNI | NULL    |       |
| phone     | varchar(15)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

5. Write the SQL command to create a table Borrow_Records with:
   - record_id (Primary Key, INT)
   - member_id (Foreign Key referencing Members)
   - book_id (Foreign Key referencing Books)
   - borrow_date (DATE)
   - return_date (DATE)

```
mysql> create table Borrow_Records(
    -> record_id int primary key auto_increment,
    -> member_id int,
    -> book_id int,
    -> borrow_date date,
    -> return_date date,
    -> constraint fk_member_id foreign key (member_id) references Members(member_id) on delete cascade,
    -> constraint fk_book_id foreign key (book_id) references Books(book_id) on delete cascade);
Query OK, 0 rows affected (0.06 sec)

mysql> desc borrow_records;
+-------------+------+------+-----+---------+----------------+
| Field       | Type | Null | Key | Default | Extra          |
+-------------+------+------+-----+---------+----------------+
| record_id   | int  | NO   | PRI | NULL    | auto_increment |
| member_id   | int  | YES  | MUL | NULL    |                |
| book_id     | int  | YES  | MUL | NULL    |                |
| borrow_date | date | YES  |     | NULL    |                |
| return_date | date | YES  |     | NULL    |                |
+-------------+------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

6. Modify the Books table to add a column genre of type VARCHAR(50).

```
mysql> alter table Books
    -> add genre varchar(50);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Books;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| author_id        | int          | YES  | MUL | NULL    |       |
| published_year   | year         | YES  |     | NULL    |       |
| available_copies | int          | YES  |     | NULL    |       |
| genre            | varchar(50)  | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

7. Write the SQL command to drop the Borrow_Records table.

```
mysql> drop table Borrow_Records;
Query OK, 0 rows affected (0.01 sec)

mysql> show tables;
+----------------+
| Tables_in_lms  |
+----------------+
| authors        |
| books          |
| members        |
| temp           |
| temp_foreign   |
+----------------+
5 rows in set (0.01 sec)
```

8. Insert 3 records into the Authors table.

```
mysql> insert into Authors (author_id,name, country) values (1,'Rohit','India'),(2,'Nikhil','India'),(3,'Suraj','India');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from authors;
+-----------+--------+---------+
| author_id | name   | country |
+-----------+--------+---------+
|         1 | Rohit  | India   |
|         2 | Nikhil | India   |
|         3 | Suraj  | India   |
+-----------+--------+---------+
3 rows in set (0.00 sec)
```

9. Insert 5 books into the Books table.

```
mysql> insert into Books (book_id,title,author_id,published_year,available_copies,genre)
    -> values (1,'Data System',1,2026,100,'Tech'),(2,'DSA',3,2025,100,'Tech'),(3,'Java',2,2022,50,'Tech'),(4,'Cpp',1,2020,80,'Tech'),
(5,'Web design',3,2016,100,'Tech');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> d
    -> ^C
mysql> select * from Books;
+---------+-------------+-----------+----------------+------------------+-------+
| book_id | title       | author_id | published_year | available_copies | genre |
+---------+-------------+-----------+----------------+------------------+-------+
|       1 | Data System |         1 |           2026 |              100 | Tech  |
|       2 | DSA         |         3 |           2025 |              100 | Tech  |
|       3 | Java        |         2 |           2022 |               50 | Tech  |
|       4 | Cpp         |         1 |           2020 |               80 | Tech  |
|       5 | Web design  |         3 |           2016 |              100 | Tech  |
+---------+-------------+-----------+----------------+------------------+-------+
5 rows in set (0.00 sec)
```

10. Insert 3 members into the Members table.

```
mysql> insert into members(member_id,name ,email,phone)
    -> values (1,'Pranit','pranit@gmail.com','12345678');
Query OK, 1 row affected (0.01 sec)

mysql> insert into members(member_id,name ,email,phone)
    -> values (2,'Amol','amol@gmail.com','13243546'),
    -> (3,'NIkhil','nikhil@gmail.com','24537648');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> select * from members;
+-----------+--------+-----------------+----------+
| member_id | name   | email           | phone    |
+-----------+--------+-----------------+----------+
|         1 | Pranit | pranit@gmail.com | 12345678 |
|         2 | Amol   | amol@gmail.com   | 13243546 |
|         3 | NIkhil | nikhil@gmail.com | 24537648 |
+-----------+--------+-----------------+----------+
3 rows in set (0.00 sec)
```

11. Insert 4 borrow records into the Borrow_Records table.

```
mysql> insert into borrow_records(record_id,member_id,book_id,borrow_date,return_date)
    -> values(1,2,1,'2026-10-09','2026-10-15'),
    -> (2,1,3,'2026-10-04','2026-10-30'),
    -> (3,3,2,'2025-08-12','2026-01-01'),
    -> (4,2,1,'2026-08-10','2026-11-19');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * from borrow_records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|         1 |         2 |       1 | 2026-10-09  | 2026-10-15  |
|         2 |         1 |       3 | 2026-10-04  | 2026-10-30  |
|         3 |         3 |       2 | 2025-08-12  | 2026-01-01  |
|         4 |         2 |       1 | 2026-08-10  | 2026-11-19  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)
```

12. Write an SQL query to select all books where published_year is after 2015.

```
mysql> select * from books where published_year > 2015;
+---------+-------------+-----------+----------------+------------------+-------+
| book_id | title       | author_id | published_year | available_copies | genre |
+---------+-------------+-----------+----------------+------------------+-------+
|       1 | Data System |         1 |           2026 |              100 | Tech  |
|       2 | DSA         |         3 |           2025 |              100 | Tech  |
|       3 | Java        |         2 |           2022 |               50 | Tech  |
|       4 | Cpp         |         1 |           2020 |               80 | Tech  |
|       5 | Web design  |         3 |           2016 |              100 | Tech  |
+---------+-------------+-----------+----------------+------------------+-------+
5 rows in set (0.00 sec)
```

13. Write a SQL query to create a foreign key & primary key relationship between two tables.

```
mysql> create table temp(temp_id int primary key, name varchar(10));
Query OK, 0 rows affected (0.02 sec)

mysql> desc temp;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| temp_id | int         | NO   | PRI | NULL    |       |
| name    | varchar(10) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)

mysql> create table temp_foreign (temp_fId int primary key,name varchar(10),temp_id int, constraint fk_temp_key foreign key (temp_id)
 references temp(temp_id) on delete cascade);
Query OK, 0 rows affected (0.04 sec)

mysql> desc temp_foreign;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| temp_fId | int         | NO   | PRI | NULL    |       |
| name     | varchar(10) | YES  |     | NULL    |       |
| temp_id  | int         | YES  | MUL | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

14. Write an SQL query to find all members who have borrowed the book with title 'Database Systems'.

```
mysql> select m.*
    -> from members m
    -> join borrow_records br
    -> on m.member_id = br.member_id
    -> join books b
    -> on br.book_id = b.book_id
    -> where b.title = 'Data System';
+-----------+------+-----------------+-----------+
| member_id | name | email           | phone     |
+-----------+------+-----------------+-----------+
|         2 | Amol | amol@gmail.com | 13243546 |
|         2 | Amol | amol@gmail.com | 13243546 |
+-----------+------+-----------------+-----------+
2 rows in set (0.00 sec)
```

15. Update the available_copies column of a specific book (choose any book) by reducing it by 1 after it is borrowed.

```
mysql> update books set available_copies = available_copies - 1 where book_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from books;
+---------+-------------+-----------+----------------+------------------+-------+
| book_id | title       | author_id | published_year | available_copies | genre |
+---------+-------------+-----------+----------------+------------------+-------+
|       1 | Data System |         1 |           2026 |               99 | Tech  |
|       2 | DSA         |         3 |           2025 |              100 | Tech  |
|       3 | Java        |         2 |           2022 |               50 | Tech  |
|       4 | Cpp         |         1 |           2020 |               80 | Tech  |
|       5 | Web design  |         3 |           2016 |              100 | Tech  |
+---------+-------------+-----------+----------------+------------------+-------+
5 rows in set (0.00 sec)
```

16. Delete a record from Members where member_id = 3.

```
mysql> delete from Members where member_id = 3;
Query OK, 1 row affected (0.01 sec)

mysql> select * from members;
+-----------+--------+-----------------+-----------+
| member_id | name   | email           | phone     |
+-----------+--------+-----------------+-----------+
|         1 | Pranit | pranit@gmail.com | 12345678 |
|         2 | Amol   | amol@gmail.com   | 13243546 |
+-----------+--------+-----------------+-----------+
2 rows in set (0.00 sec)
```

17. Update a Book name record from Book table with id = 1.

```
mysql> update books set title = 'Data' where book_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from books;
+---------+------------+-----------+----------------+------------------+-------+
| book_id | title      | author_id | published_year | available_copies | genre |
+---------+------------+-----------+----------------+------------------+-------+
|       1 | Data       |         1 |           2026 |               99 | Tech  |
|       2 | DSA        |         3 |           2025 |              100 | Tech  |
|       3 | Java       |         2 |           2022 |               50 | Tech  |
|       4 | Cpp        |         1 |           2020 |               80 | Tech  |
|       5 | Web design |         3 |           2016 |              100 | Tech  |
+---------+------------+-----------+----------------+------------------+-------+
5 rows in set (0.00 sec)
```

18. Write an SQL query to list all books along with their authors' names.

```
mysql> select books.title, Authors.name
    -> from books
    -> join Authors
    -> on books.author_id = Authors.author_id;
+------------+--------+
| title      | name   |
+------------+--------+
| Data       | Rohit  |
| Cpp        | Rohit  |
| Java       | Nikhil |
| DSA        | Suraj  |
| Web design | Suraj  |
+------------+--------+
5 rows in set (0.00 sec)
```

19. Write an SQL query to delete all books from the Books table where the published_year is before 2000.

```
mysql> delete from books where published_year < 2000;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from books;
+---------+------------+-----------+----------------+------------------+-------+
| book_id | title      | author_id | published_year | available_copies | genre |
+---------+------------+-----------+----------------+------------------+-------+
|       1 | Data       |         1 |           2026 |               99 | Tech  |
|       2 | DSA        |         3 |           2025 |              100 | Tech  |
|       3 | Java       |         2 |           2022 |               50 | Tech  |
|       4 | Cpp        |         1 |           2020 |               80 | Tech  |
|       5 | Web design |         3 |           2016 |              100 | Tech  |
+---------+------------+-----------+----------------+------------------+-------+
5 rows in set (0.00 sec)
```

20. Write an SQL query to **find all books that are never borrowed (i.e., no records in `Borrow_Records`).**

```
mysql> select books.book_id,books.title from books
    -> left join borrow_records
    -> on books.book_id =  borrow_records.book_id
    -> where borrow_records.book_id is null;
+---------+------------+
| book_id | title      |
+---------+------------+
|       2 | DSA        |
|       4 | Cpp        |
|       5 | Web design |
+---------+------------+
3 rows in set (0.00 sec)
```

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